

BSc(Surveying) | RECO40II Dissertation

Mapping the Barriers Network to Drone Adoption
in Indoor Building Inspections in Hong Kong:
An Interpretive Structural Modelling (ISM) Study

STUDENT: Chan Chun Hei

SUPERVISOR: Prof. Frank Xue



Major Findings

1. The barriers to drone adoption are highly interrelated and cannot be addressed in isolation.
2. Technical barriers are the root cause preventing drone adoption.
3. Promoting drone adoption requires technological advancement, subsidies, training, and infrastructure upgrades.
4. The current feasible scope of indoor drone inspection is limited to lift shafts, atriums and high ceilings.

Abstract

Hong Kong is home to over 11,000 buildings that have exceeded 50 years of age. This ageing stock of buildings creates strong demand for safer, faster and more cost-effective building inspections. Traditional methods for inspecting hard-to-reach areas and voids are time-consuming, costly and prone to safety risks due to working at height. While Unmanned Aerial Systems have been widely adopted in outdoor façade inspections, their use in indoor building environments remains largely underdeveloped and understudied, particularly in Hong Kong.

A mixed-method approach was adopted. First, a systematic literature review of 32 studies identified 20 potential barriers across five categories. These barriers were evaluated by 52 building professionals using a Likert-scale questionnaire, narrowing the list to 12 critical barriers. Semi-structured interviews with three experts were then conducted to establish the directional relationships among these barriers, which were subsequently modelled using Interpretive Structural Modelling (ISM) and MICMAC analysis.

The findings reveal that the 12 barriers are highly interconnected. Technical and Operational barriers sit at the base of the ISM hierarchy as root barriers, influencing all other barriers in the network. In contrast, Safety barriers sit at the top as the most dependent outcomes, driven by unresolved technical, data, and financial barriers. Data processing and financial barriers occupy the middle of the hierarchy as linkage barriers, amplifying the influence of root barriers on safety outcomes.

These findings confirm that indoor drone adoption cannot be achieved by addressing any single barrier in isolation. Collective action from multiple stakeholders is required. It is recommended that drone manufacturers prioritise autonomous navigation and improved battery systems, while the Construction Industry Council (CIC) develops standardised indoor drone operator training aligned with MBIS requirements. Additionally, facilities managers should begin structured pilot programmes in lift shafts and high-ceiling voids, which are the most feasible near-term applications under current technology.

